

KNOWLEDGE REPRESENTATION

Lecture 4: Actions with Default Effects, Reasoning about Scenarios

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Certain and default effects

- Actions we have considered recently, deterministic or non-deterministic, have a common feature – their effects were **certain**. No preferences (expectations) were involved.
- In real-life problems actions usually have also **default** (expected, preferred) effects. They are not certain and in some (atypical) cases they need not occur.

Example: Two shooting actions

- ① Shooting the gun (by Bill) **always** makes it unloaded and, if the gun was loaded before the action, makes Fred dead.
- ② Shooting the gun (by Jim) **always** makes it unloaded and, if the gun was loaded before, it **typically** makes Fred dead.

Default effects

General assumption

Unless otherwise is known, assume that typical effects occur.

Qualification problem

Default effects might be also viewed from the standpoint of the qualification problem: since it is in general impossible to enumerate all conditions under which the action is performed successfully and brings given results, we state that it *usually* has these results under specified (not necessary completely) conditions.

Basic assumptions

Dynamic systems with defaults

- Inertia law.
- Complete information about all actions and all fluents.
- Non-determinism.
- Side effects (ramifications) of actions are admitted.
- Action can be unexecutable in some states.
- Actions may have *certain* or *typical* (preferable) effects.

Action Language $\mathcal{AD}$

To represent this class of dynamic systems we will use action languages of the class $\mathcal{AD}$.

Action Language $\mathcal{AD}$ – Statements

Signature

A **signature** is a pair $\Upsilon = (\mathcal{F}, \mathcal{Ac})$ defined as before.

Statements

The set of statements in $\mathcal{AD}$ is that of $\mathcal{AR}$ extended by the **typical effect statement** of the form:

A typically causes α if π

Intuitively: the action A , whenever performed in a state satisfying π , **typically** leads to a state satisfying α . The effect α is **preferred** – in contrast to nondeterministic effects, where none result is preferred.

E.g. SHOOT **typically causes** $\neg$ alive **if** loaded.

State

A **state** is a mapping $\sigma : \mathcal{F} \rightarrow \{0, 1\}$.

As before, $\sigma \models \alpha$ means that α holds in σ (or σ is a model of α).

Structure

A **structure** for $\mathcal{AD}$ is a tuple $S = (\Sigma, \sigma_0, ResAb, ResN)$ where

- Σ is a set of states;
- $\sigma_0 \in \Sigma$ is the initial state;
- $ResAb, ResN : \mathcal{Ac} \times \Sigma \rightarrow 2^\Sigma$ are transition functions; $ResAb$ and $ResN$ are an **atypical transition** and a **typical transition** functions, respectively.

Action Language $\mathcal{AD}$ – Semantics (cont.)

Let D be an action domain in $\mathcal{AD}$ and let $S = (\Sigma, \sigma_0, ResAb, ResN)$ be a structure for $\mathcal{AD}$.

States of D

A state σ is a **state of D** iff $\sigma \models \alpha$ for every constraint (**always** α) in D .

Ψ mapping

Let $A \in \mathcal{Ac}$ and let σ . Put $Res(A, \sigma) = ResAb(A, \sigma) \cup ResN(A, \sigma)$. Define $\Psi_S : \mathcal{Ac}^* \times \Sigma \rightarrow \Sigma$ as follows:

- $\Psi_S(\varepsilon, \sigma) = \sigma$;
- if $\Psi_S((A_1, \dots, A_n), \sigma)$ is defined, then
 $\Psi_S((A_1, \dots, A_n), \sigma) \in Res(A_n, \Psi_S((A_1, \dots, A_{n-1}), \sigma))$.

Satisfiability of value statements

- A value statement α **after** $A_1, \dots, A_n$ is **true** in S iff for **every** mapping Ψ_S defined before, if $\Psi_S((A_1, \dots, A_n), \sigma_0)$ is defined, then $\Psi_S((A_1, \dots, A_n), \sigma_0) \models \alpha$.
- An observation statement **observable** α **after** $A_1, \dots, A_n$ is **true** in S iff for **some** mapping Ψ_S ,
 - $\Psi_S((A_1, \dots, A_n), \sigma_0)$ is defined;
 - $\Psi_S((A_1, \dots, A_n), \sigma_0) \models \alpha$.

Auxiliary functions

- $Res_0 : \mathcal{Ac} \times \Sigma \rightarrow 2^\Sigma$ is defined wrt **effect statements**: for every $A \in \mathcal{Ac}$ and for every state $\sigma \in \Sigma$, $Res_0(A, \sigma)$ is the set of all states $\sigma' \in \Sigma$ satisfying:

$$(A \text{ **causes** } \alpha \text{ **if** } \pi) \in D \quad \& \quad (\sigma \models \pi) \implies \sigma' \models \alpha.$$

- $Res^- : \mathcal{Ac} \times \Sigma \rightarrow 2^\Sigma$ is defined wrt minimization of changes, i.e., for every action $A \in \mathcal{Ac}$ and for every state $\sigma \in \Sigma$, $Res^-(A, \sigma)$ is the set of all states $\sigma' \in Res_0(A, \sigma)$ for which the sets $New(A, \sigma, \sigma')$ are minimal (wrt set inclusion).
 - $Res_0^+ : \mathcal{Ac} \times \Sigma \rightarrow 2^\Sigma$ is defined as follows: for any $A \in \mathcal{Ac}$ and for any $\sigma \in \Sigma$, $Res_0^+(A, \sigma)$ is the set of states $\sigma' \in Res_0(A, \sigma)$ such that
- $$(A \text{ **typically causes** } \alpha \text{ **if** } \pi) \in D \quad \& \quad (\sigma \models \pi) \implies (\sigma' \models \alpha).$$

$Res_0^+(A, \sigma)$ contains states where both certain and typical effects of A hold.

Model

Let D be an action domain of $\mathcal{AD}$. a structure $S = (\Sigma, \sigma_0, ResAb, ResN)$ is a **model** of D iff

- 1 Σ is the set of all states of D ;
- 2 All value statements and all observation statements in D are true in S ;
- 3 For every $A \in \mathcal{Ac}$ and for every $\sigma \in \Sigma$,

$$ResN(A, \sigma) = \{\sigma' \in Res_0^+(A, \sigma) : New(A, \sigma, \sigma') \text{ are minimal}\}$$
$$ResAb(A, \sigma) = Res^-(A, \sigma) \setminus ResN(A, \sigma).$$

$ResN$ is the set of all expected states (wrt minimal changes) satisfying both certain and typical effects, while $ResAb$ is the set of expected states, where only certain effects hold.

Example: Modification of YSP

Example 4.1

initially $alive \wedge \neg loaded$;
LOAD *causes* $loaded$;
SHOOT *causes* $\neg loaded$;
SHOOT *typically causes* $\neg alive$ *if* $loaded$.

$\Sigma = \{\sigma_0, \sigma_1, \sigma_2, \sigma_3\}$ where

$\sigma_0 = \{alive, \neg loaded\}$ $\sigma_2 = \{\neg alive, \neg loaded\}$
 $\sigma_1 = \{alive, loaded\}$ $\sigma_3 = \{\neg alive, loaded\}$.

$Res_0(\text{SHOOT}, \sigma_0) = \{\sigma_0, \sigma_2\}$

$Res^-(\text{SHOOT}, \sigma_0) = \{\sigma_0\}$

$Res^+(\text{SHOOT}, \sigma_0) = \{\sigma_0, \sigma_2\}$

$ResN(\text{SHOOT}, \sigma_0) = \{\sigma_0\}$

$ResAb(\text{SHOOT}, \sigma_0) = \emptyset$.

Modification of YSP (cont.)

Example 4.1

$$\begin{aligned}\sigma_0 &= \{alive, \neg loaded\} & \sigma_2 &= \{\neg alive, \neg loaded\} \\ \sigma_1 &= \{alive, loaded\} & \sigma_3 &= \{\neg alive, loaded\}.\end{aligned}$$

$$Res_0(\text{LOAD}, \sigma_0) = \{\sigma_1, \sigma_3\}$$

$$Res^-(\text{LOAD}, \sigma_0) = \{\sigma_1\}$$

$$Res_0^+(\text{LOAD}, \sigma_0) = \{\sigma_1, \sigma_3\}$$

$$ResN(\text{LOAD}, \sigma_0) = \{\sigma_1\}$$

$$ResAb(\text{LOAD}, \sigma_0) = \emptyset.$$

Note that whenever an action has certain effects only, they are viewed as **typical**.

Modification of YSP (cont.)

Example 4.1

$$\begin{aligned}\sigma_0 &= \{alive, \neg loaded\} & \sigma_2 &= \{\neg alive, \neg loaded\} \\ \sigma_1 &= \{alive, loaded\} & \sigma_3 &= \{\neg alive, loaded\}.\end{aligned}$$

$$Res_0(\text{LOAD}, \sigma_1) = \{\sigma_1, \sigma_3\}$$

$$Res^-(\text{LOAD}, \sigma_1) = \{\sigma_1\}$$

$$Res_0^+(\text{LOAD}, \sigma_1) = \{\sigma_1, \sigma_3\}$$

$$ResN(\text{LOAD}, \sigma_1) = \{\sigma_1\}$$

$$ResAb(\text{LOAD}, \sigma_0) = \emptyset.$$

$$Res_0(\text{SHOOT}, \sigma_1) = \{\sigma_0, \sigma_2\}$$

$$Res^-(\text{SHOOT}, \sigma_1) = \{\sigma_0\}$$

$$Res_0^+(\text{SHOOT}, \sigma_1) = \{\sigma_2\}$$

$$ResN(\text{SHOOT}, \sigma_1) = \{\sigma_2\}$$

$$ResAb(\text{SHOOT}, \sigma_0) = \{\sigma_0\}.$$

Modification of YSP (cont.)

Example 4.1

$$\begin{aligned}\sigma_0 &= \{alive, \neg loaded\} & \sigma_2 &= \{\neg alive, \neg loaded\} \\ \sigma_1 &= \{alive, loaded\} & \sigma_3 &= \{\neg alive, loaded\}.\end{aligned}$$

$$Res_0(\text{LOAD}, \sigma_2) = \{\sigma_1, \sigma_3\}$$

$$Res^-(\text{LOAD}, \sigma_2) = \{\sigma_3\}$$

$$Res_0^+(\text{LOAD}, \sigma_2) = \{\sigma_1, \sigma_3\}$$

$$ResN(\text{LOAD}, \sigma_2) = \{\sigma_3\}$$

$$ResAb(\text{LOAD}, \sigma_2) = \emptyset.$$

$$Res_0(\text{SHOOT}, \sigma_2) = \{\sigma_0, \sigma_2\}$$

$$Res^-(\text{SHOOT}, \sigma_2) = \{\sigma_2\}$$

$$Res_0^+(\text{SHOOT}, \sigma_2) = \{\sigma_0, \sigma_2\}$$

$$ResN(\text{SHOOT}, \sigma_2) = \{\sigma_2\}$$

$$ResAb(\text{SHOOT}, \sigma_2) = \emptyset.$$

Modification of YSP (cont.)

Example 4.1

$$\begin{aligned}\sigma_0 &= \{alive, \neg loaded\} & \sigma_2 &= \{\neg alive, \neg loaded\} \\ \sigma_1 &= \{alive, loaded\} & \sigma_3 &= \{\neg alive, loaded\}.\end{aligned}$$

$$Res_0(\text{LOAD}, \sigma_3) = \{\sigma_1, \sigma_3\}$$

$$Res^-(\text{LOAD}, \sigma_3) = \{\sigma_3\}$$

$$Res_0^+(\text{LOAD}, \sigma_3) = \{\sigma_1, \sigma_3\}$$

$$ResN(\text{LOAD}, \sigma_3) = \{\sigma_3\}$$

$$ResAb(\text{LOAD}, \sigma_3) = \emptyset.$$

$$Res_0(\text{SHOOT}, \sigma_3) = \{\sigma_0, \sigma_2\}$$

$$Res^-(\text{SHOOT}, \sigma_3) = \{\sigma_2\}$$

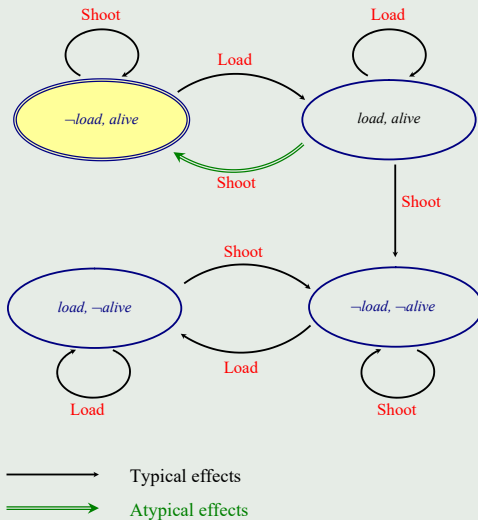
$$Res_0^+(\text{SHOOT}, \sigma_3) = \{\sigma_2\}$$

$$ResN(\text{SHOOT}, \sigma_3) = \{\sigma_2\}$$

$$ResAb(\text{SHOOT}, \sigma_3) = \emptyset.$$

Modification of YSP (cont.)

Example 4.1



Question

Is it possible that an action has only atypical effects in some state?

Example 4.2

always $f \vee g$;
A *causes* $\neg f$;
A *typically causes* $\neg g$ if f .

$\sigma_0 = \{f, g\}$	$Res_0(A, \sigma_0) = \{\sigma_2\}$	
$\sigma_1 = \{f, \neg g\}$	$Res^-(A, \sigma_0) = \{\sigma_2\}$	$ResN(A, \sigma_0) = \emptyset$
$\sigma_2 = \{\neg f, g\}$	$Res_0^+(A, \sigma_0) = \emptyset$	$ResAb(A, \sigma_0) = \{\sigma_2\}.$

$$\alpha \text{ after } A_1, \dots, A_n$$
$$\textit{typically } \alpha \text{ after } A_1, \dots, A_n$$

Intutively:

- The 1st query states that α **always** holds after performind the sequence $(A_1, \dots, A_n)$ of actions, regarless of the way the actions were executed, that is, in a typical or atypical way.
- The 2nd query says that α **typically** holds after performing the sequence $(A_1, \dots, A_n)$ of actions which is understood as **minimal atypical transitions** on paths determined by $(A_1, \dots, A_n)$.

Abnormality measure

Penalty function

Let $S = (\Sigma, \sigma_0, ResAb, ResN)$ be a structure for $\mathcal{AD}$. For any action $A \in \mathcal{Ac}$ and for any two states $\sigma, \sigma' \in \Sigma$ define

$$\kappa_S(A, \sigma, \sigma') = \begin{cases} 0 & \text{iff } \sigma' \in ResN(A, \sigma) \\ 1 & \text{iff } \sigma' \in ResAb(A, \sigma) \\ +\infty & \text{otherwise} \end{cases}$$

Generalization for paths

For any mapping Ψ_S and for any sequence $(A_1, \dots, A_n)$ of actions:

- $K_{\Psi_S}((A_1, \dots, A_n), \sigma_0) = +\infty$ iff $\Psi_S((A_1, \dots, A_n), \sigma_0)$ is undefined;
- $K_{\Psi_S}((A_1, \dots, A_n), \sigma_0) = \sum_{i=1}^n \kappa_S(A_i, \sigma_{i-1}, \sigma_i), \sigma_i = \Psi_S(A_i, \sigma_{i-1}).$

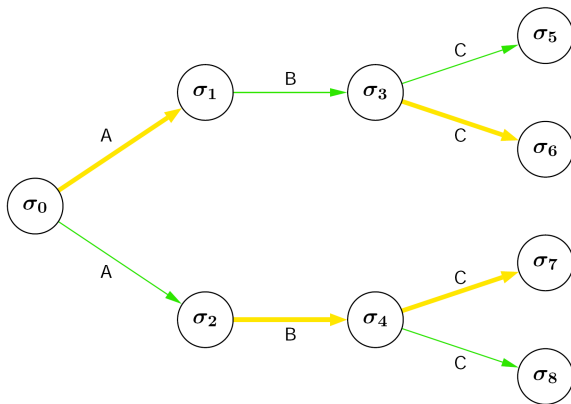
Abnormality measure (cont.)

Preferred paths

For any sequence $(A_1, \dots, A_n)$ of actions, $n \geq 1$, let $\Pi_S(A_1, \dots, A_n)$ stand for the set of all Ψ_S mappings for which $K_{\Psi_S}((A_1, \dots, A_n), \sigma_0)$ are minimal. Every $\Psi_S \in \Pi_S(A_1, \dots, A_n)$ is a **preferred path** for $(A_1, \dots, A_n)$.

Mappings Ψ_S and the set $\Pi_S(A, B, C)$

Example 4.3



Example 4.4

Consequences of D

Let D be an action domain in a language of the class $\mathcal{AD}$.

Q : α *after* $A_1, \dots, A_n$

Q is a **consequence** of D , in symbols $D \models Q$, iff for every model $S = (\Sigma, \sigma_0, ResAb, ResN)$ of D and for every mapping Ψ_S ,
 $\Psi_S((A_1, \dots, A_n), \sigma_0) \models \alpha$.

Q : α *typically after* $A_1, \dots, A_n$

Q is a **consequence** of D , in symbols $D \models Q$, iff for every model $S = (\Sigma, \sigma_0, ResAb, ResN)$ of D and for every mapping $\Psi_S^* \in \Pi_S(A_1, \dots, A_n)$, $\Psi_S^*((A_1, \dots, A_n), \sigma_0) \models \alpha$.

Example: Modification of YSP (cont.)

Example 4.1

initially $alive \wedge \neg loaded$;

LOAD *causes* loaded;

SHOOT *causes* $\neg loaded$;

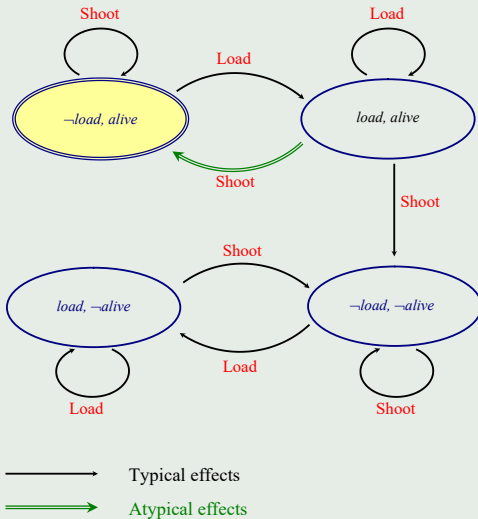
SHOOT *typically causes* $\neg alive$ *if* loaded;

Note that

$D \models \textit{typically } \neg alive \textit{ after } LOAD, SHOOT.$

Example: Modification of YSP (cont.)

Example 4.1



Final remarks

In $\mathcal{AD}$ we assume that constraints work for actions' ramifications. We can consider action languages with default effects, where constraints influence on qualifications of actions. To this end,

- Consider D^- obtained by removing all constraints from D ;
- Determine models of D^- in the sense of $\mathcal{AD}$. Let $S^- = (\Sigma^-, \sigma, ResAB^-, ResN^-)$ be such a model where σ_0 satisfies all constraints of D . Put
 - $\Sigma \subseteq \Sigma^-$ is the set of all states satisfying all constraints;
 - For any $A \in \mathcal{Ac}$ and for any $\sigma \in \Sigma$,

$$ResN(A, \sigma) = ResN^-(A, \sigma) \cap \Sigma$$

$$ResAb(A, \sigma) = ResAb^-((A, \sigma) \cap \Sigma).$$

Reasoning about Scenarios

Questions are welcome.

Basic assumptions

In contrast to previous approaches with *branching time* model, now we will consider action domains with *linear time*.

Assumptions

- ① Inertia law.
- ② Linear model of time (discrete time).
- ③ Actions with duration; during performance of the action, values of fluents changed by the actions are unknown.
- ④ Dynamic effects of actions – one action can invoke another one(s).
- ⑤ Situation can trigger actions – some states may cause executing some actions.

Action Language $\mathcal{AL}$

To represent these dynamic systems we will use action languages of the class $\mathcal{AL}$. We assume that each action is performed in 1 unit of time.

Action Language $\mathcal{AL}$ – Syntax

Signature

A **signature** is a pair $\Upsilon = (\mathcal{F}, \mathcal{Ac})$ where $\mathcal{F}$ is a set of *fluents* and $\mathcal{Ac}$ is a set of *actions*.

Formula

A **formula** is a propositional combination of fluents, that is

$$\alpha ::= f \mid \neg\alpha \mid \alpha \wedge \beta \mid \alpha \vee \beta \mid \alpha \rightarrow \beta \mid \alpha \rightarrow \beta.$$

Statements in $\mathcal{AL}$

Fluent effect statement

A causes α if π

INTUITIVE READING:

Whenever the action A starts in a state satisfying π , it leads to a state where α holds.

Action effect statement

A invokes B after d if π

INTUITIVE READING:

Whenever the action A starts in a state satisfying π , after $d \in \mathbb{N}$ units of time after terminating A the action B starts.

Statements in $\mathcal{AL}$ (cont.)

Release statement

A releases f if π

INTUITIVE READING:

Whenever the action A starts in a state satisfying π , after its execution the value of the fluent f might, but need not change.

Trigger statement

π triggers A

INTUITIVE READING:

The action A starts whenever the condition π holds.

Domain description

A finite set of statements is a *domain description*.

Action Scenarios

An **action scenario** (**scenario**, for short) is a pair $Sc = (Obs, Acs)$ where

- Obs is a set of **observations** $Obs = \{(\alpha_1, t_1), \dots, (\alpha_n, t_n)\}$, α_i is a formula and $t_i \in \mathbb{N}$ is a timepoint, $i = 1, \dots, n$;
- Acs is a set of **action occurrences** $Acs = \{(A_1, t_1), \dots, (A_k, t_k)\}$, A_j is an action and $t_j \in \mathbb{N}$ is a timepoint, $j = 1, \dots, k$.

Structure

A **structure** for a language of the class $\mathcal{AL}$ is a triple $S = (H, O, E)$ such that

- $H : \mathcal{F} \times \mathbb{N} \rightarrow \{0, 1\}$ is a **history function**;
- $O : \mathcal{Ac} \times \mathbb{N} \rightarrow 2^{\mathcal{F}}$ is an **occlusion function**; for every $A \in \mathcal{Ac}$ and for every timepoint $t \in \mathbb{N}$, $O(A, t)$ is the set of fluents under influence of executing A from timepoint $t - 1$ to t ;
- $E \subseteq \mathcal{Ac} \times \mathbb{N}$ is an **action occurrences relation**; if $(A, t) \in E$, then A occurs at timepoint t .

For all actions $A, B \in \mathcal{Ac}$ and for every $t \in \mathbb{N}$,

$$(A, t) \in E \ \& \ (B, t) \in E \implies A = B \quad (1)$$

The condition (1) guarantees that at most one action is executed at a time.

History function

For any structure $S = (H, O, E)$, the history function H is extended to the set of all formulas according to rules well-known in propositional logic, that is, for every timepoint $T \in \mathbb{N}$,

$$H^*(f, t) = H(f, t) \text{ for every } f \in \mathcal{F};$$

$$H^*(\neg\alpha, t) = 1 - H^*(\alpha, t);$$

$$H^*(\alpha \wedge \beta, t) = \min(H^*(\alpha, t), H^*(\beta, t));$$

$$H^*(\alpha \vee \beta, t) = \max(H^*(\alpha, t), H^*(\beta, t));$$

$$H^*(\alpha \rightarrow \beta, t) = \begin{cases} 0 & \text{if } H^*(\alpha, t) = 1 \text{ \& } H^*(\beta, t) = 0 \\ 1 & \text{otherwise} \end{cases}$$

$$H^*(\alpha \leftrightarrow \beta, t) = \begin{cases} 1 & \text{if } H^*(\alpha, t) = H^*(\beta, t) \\ 0 & \text{otherwise} \end{cases}$$

Structure for Sc wrt D

Let D be a domain description in a language of the class $\mathcal{AL}$, let $S = (H, O, E)$ be a structure for *mathcal{AL}*, and let $Sc = (Obs, Acs)$ be a scenario. A structure S is a **structure for Sc wrt D** iff

- for every observation $(\alpha, t) \in Obs$, $H(\alpha, t) = 1$;
- $Acs \subseteq E$.

Semantics of $\mathcal{AL}$ (cont.)

Denote: $fl(\alpha)$ – the set of fluents occurring on α .

Structure for Sc wrt D

- for every statement $(A \text{ **causes** } \alpha \text{ **if** } \pi) \in D$ and for every timepoint $t \in \mathbb{N}$, if $H(\pi, t) = 1$ and $(A, t) \in E$, then $H(\alpha, t + 1) = 1$ and $fl(\alpha) \subseteq O(A, t + 1)$;
- for every statement $(A \text{ **releases** } f \text{ **if** } \pi) \in D$ and for every timepoint $t \in \mathbb{N}$, if $H(\pi, t) = 1$ and $(A, t) \in E$, then $f \in O(A, t + 1)$;
- for every statement $(\pi \text{ **triggers** } A) \in D$ and for every timepoint $t \in \mathbb{N}$, if $H(\pi, t) = 1$, then $(A, t) \in e$;
- for every statement $(A \text{ **invokes** } B \text{ **afterd** **if** } \pi) \in D$ and for every timepoint $t \in \mathbb{N}$, if $H(\pi, t) = 1$ and $(A, t) \in E$, then $(B, t + 1 + d) \in E$.

Semantics of $\mathcal{AL}$ (cont.)

Remark 4.1

Note that

- Any change (in fluents' values) are allowed **only** in occlusion regions;
- Consequently, we will be interested in structures $S = (H, O, E)$ for $Sc = (Obs, Acs)$ wrt D which occlusion functions O determine the smallest occlusion regions.

Let $O_1, O_2 : X \rightarrow 2^Y$. We write

- $O_1 \preceq O_2$ iff $O_1(x) \subseteq O_2(x)$ for every $x \in X$;
- $O_1 \prec O_2$ iff $O_1 \preceq O_2$ and $O_1 \neq O_2$.

O-minimal structures

Let $S = (H, O, E)$ be a structure for a scenario $Sc = (Obs, Acs)$ wrt a domain description D . We say that S is O -minimal iff there is no structure $S' = (H', O', E')$ for Sc wrt D such that $O' \prec O$.

Semantics of $\mathcal{AL}$ (cont.)

Model

An O -minimal structure $S = (H, O, E)$ for $Sc = (Obs, Acs)$ wrt a domain description D is a **model of Sc wrt D** iff

(M1) for every timepoint $t \in \mathbb{N}$,

$$\{f \in \mathcal{F} : H(f, t) \neq H(f, t+1)\} \subseteq O(A, t+1)$$

for some action $A \in \mathcal{A}_c$;

(M2) there is no O -minimal structure $S = (H', O', E')$ for Sc wrt D satisfying **(M1)** such that $E' \subseteq E$.

Consistency

A scenario Sc is **consistent** wrt D iff there is some model S of sc wrt D ; otherwise it is called **inconsistent**.

Example 4.10

Let D be a domain description with two actions A and B and let a scenario $Sc = (Obs, Acs)$ be given as

- $Obs = \emptyset$;
- $Acs = \{(A, 1), (B, 1)\}$.

Clearly, Sc is inconsistent wrt any domain description D .

Example 4.11

Consider the following domain description D and two scenarios $Sc_1 = (Obs_1, Acs_1)$ and $Sc_2 = (Obs_2, Acs_2)$ given by:

- | | |
|--|----------------------------------|
| A causes f ; | • $Obs_1 = Obs_2 = \emptyset$ |
| B causes $\neg f$; | • $Acs_1 = \{(A, 1), (B, 3)\}$ |
| C causes g ; | • $Acs_2 = \{(A, 1), (B, 2)\}$. |
| A invokes C after 1. | |

For any $S = (H, 0, E)$ for Sc_1 wrt D , we have $(A, 1), (B, 3) \in E$. Since $(C, 3) \in E$, Sc_1 is inconsistent wrt D .

For any $S = (H, 0, E)$ for Sc_2 wrt D , $(A, 1), (B, 2), (C, 3) \in E$. Sc_2 is consistent wrt D .

Query

A **query** is an expression of the form:

necessary α at t when Sc
possibly α at t when Sc
necessary A at t when Sc
possibly A at t when Sc .

Intuitively,

The first two queries state that the condition α always/ever holds at timepoint t when the scenario Sc is carrying out, whereas the last two queries say that the action A must/may be executed at timepoint t when the scenario Sc is carrying out.

Consequences

Let Sc be a scenario and let D be a domain description. A query Q is a **consequence of Sc wrt D** , in symbols $Sc, D \models Q$, iff

- Q : **necessary** α **at** t **when** Sc
for every model $S = (H, O, E)$ of Sc wrt D , $H(\alpha, t) = 1$
- Q : **possibly** α **at** t **when** Sc
there is some model $S = (H, O, E)$ of Sc wrt D , where $H(\alpha, t) = 1$
- Q : **necessary** A **at** t **when** Sc
for every model $S = (H, O, E)$ of Sc wrt D , $(A, t) \in E$
- Q : **possibly** A **at** t **when** Sc
there is some model $S = (H, O, E)$ of Sc wrt D , where $(A, t) \in E$.

Example: Yale Shooting Problem

Example 4.12

Consider the following domain description D :

LOAD *causes* loaded;
LOAD *invokes* ESCAPE;
ESCAPE *releases* hidden;
SHOOT *causes* $\neg$ loaded;
SHOOT *causes* $\neg$ alive *if* loaded $\wedge$ $\neg$ hidden.

and the scenario $Sc = (OBS, ACS)$ where

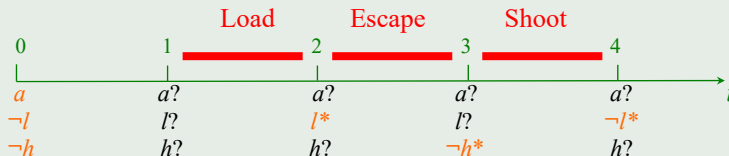
$OBS = \{(alive, \wedge \neg loaded \wedge \neg hidden, 0)\}$
 $ACS = \{(LOAD, 1), (SHOOT, 3)\}.$

Example: Yale Shooting Problem

Example 4.12 (cont.)

There are two main classes of structures for Sc wrt D , namely

Class 1:

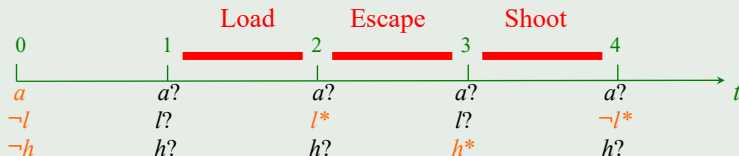


- Occlusion regions:
 - $\{l\} \subseteq Occlude(\text{LOAD}, 2)$
 - $\{h\} \subseteq Occlude(\text{ESCAPE}, 3)$
 - $\{l\} \subseteq Occlude(\text{SHOOT}, 4)$
- Occurrences of actions: $\{(\text{LOAD}, 1), (\text{ESCAPE}, 2), (\text{SHOOT}, 3)\} \subseteq E$.

Example: Yale Shooting Problem

Example 4.12 (cont.)

Class 2:



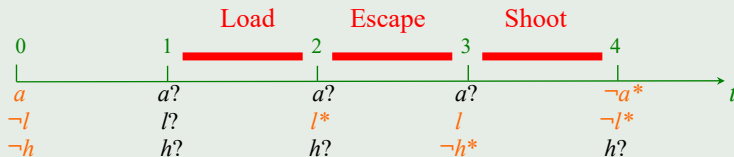
- $\{l\} \subseteq \text{Occlude}(\text{LOAD}, 2)$
- Occlusion regions: $\{h\} \subseteq \text{Occlude}(\text{ESCAPE}, 3)$
 $\{l\} \subseteq \text{Occlude}(\text{SHOOT}, 4)$
- Occurrences of actions: $\{(\text{LOAD}, 1), (\text{ESCAPE}, 2), (\text{SHOOT}, 3)\} \subseteq E$.

Example: Yale Shooting Problem

Example 4.12 (cont.)

Class 2 has two subclasses:

Subclass 1:

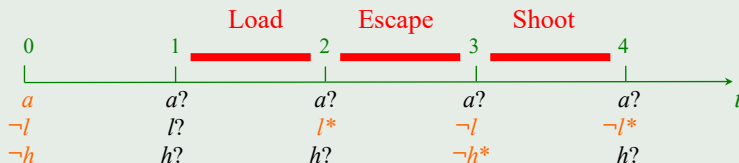


- $\{l\} \subseteq \text{Occlude}(\text{LOAD}, 2)$
- Occlusion regions: $\{h\} \subseteq \text{Occlude}(\text{ESCAPE}, 3)$
 $\{l\} \subseteq \text{Occlude}(\text{SHOOT}, 4)$
- Occurrences of actions: $\{(\text{LOAD}, 1), (\text{ESCAPE}, 2), (\text{SHOOT}, 3)\} \subseteq E$.

Example: Yale Shooting Problem

Example 4.12 (cont.)

Subclass 2:



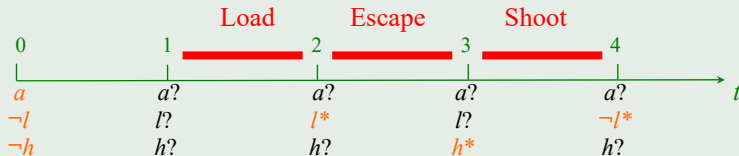
- Occlusion regions:
 $\{l\} \subseteq \text{Occlude}(\text{LOAD}, 2)$
 $\{h\} \subseteq \text{Occlude}(\text{ESCAPE}, 3)$
 $\{l\} \subseteq \text{Occlude}(\text{SHOOT}, 4)$
- Occurrences of actions: $\{(\text{LOAD}, 1), (\text{ESCAPE}, 2), (\text{SHOOT}, 3)\} \subseteq E$.

Example: Yale Shooting Problem

Example 4.12 (cont.)

Two main classes of O -minimal structures for Sc wrt D :

Class 1:



$$\{l\} = Occlude(\text{LOAD}, 2)$$

- Occlusion regions: $\{h\} = Occlude(\text{ESCAPE}, 3)$

$$\{l\} = Occlude(\text{SHOOT}, 4)$$

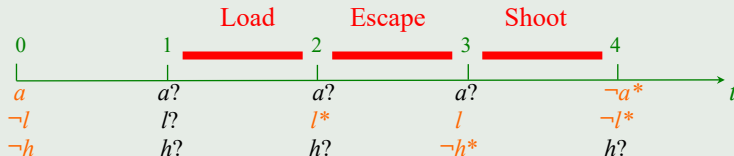
- Occurrences of actions: $\{(\text{LOAD}, 1), (\text{ESCAPE}, 2), (\text{SHOOT}, 3)\} \subseteq E$.

Example: Yale Shooting Problem

Example 4.12 (cont.)

And two subclasses of the 2nd class:

1st subclass:



$$\{l\} = \text{Occlude}(\text{LOAD}, 2)$$

- Occlusion regions: $\{h\} = \text{Occlude}(\text{ESCAPE}, 3)$

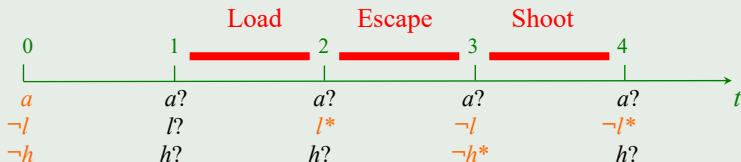
$$\{l\} = \text{Occlude}(\text{SHOOT}, 4)$$

- Occurrences of actions: $\{(\text{LOAD}, 1), (\text{ESCAPE}, 2), (\text{SHOOT}, 3)\} \subseteq E$.

Example: Yale Shooting Problem

Example 4.12 (cont.)

2nd subclass:



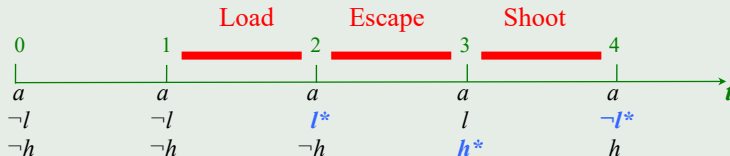
- $\{l\} = \text{Occlude}(\text{LOAD}, 2)$
- Occlusion regions: $\{h\} = \text{Occlude}(\text{ESCAPE}, 3)$
 $\{l\} = \text{Occlude}(\text{SHOOT}, 4)$
- Occurrences of actions: $\{(\text{LOAD}, 1), (\text{ESCAPE}, 2), (\text{SHOOT}, 3)\} \subseteq E$.

Example: Yale Shooting Problem

Example 4.12 (cont.)

There are two models of Sc wrt D :

1st model S_1 :



$$\{l\} = Occlude(LOAD, 2)$$

- Occlusion regions: $\{h\} = Occlude(ESCAPE, 3)$

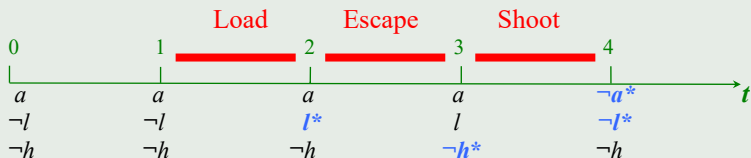
$$\{l\} = Occlude(SHOOT, 4)$$

- Occurrences of actions: $\{(LOAD, 1), (ESCAPE, 2), (SHOOT, 3)\} = E$.

Example: Yale Shooting Problem

Example 4.12 (cont.)

2nd model S_2 :

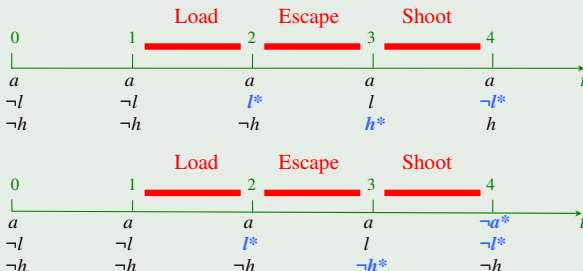


- Occlusion regions:
 $\{l\} = \text{Occlude}(\text{LOAD}, 2)$
 $\{h\} = \text{Occlude}(\text{ESCAPE}, 3)$
 $\{l\} = \text{Occlude}(\text{SHOOT}, 4)$
- Occurrences of actions: $\{(\text{LOAD}, 1), (\text{ESCAPE}, 2), (\text{SHOOT}, 3)\} = E$.

Example: Yale Shooting Problem

Example 4.12 (cont.)

Models of Sc wrt D :



- $Sc, D \models \text{necessary } \neg \text{loaded at } t \text{ when } Sc \text{ for } t \geq 4$
- $Sc, D \models \text{necessary ESCAPE at } 2 \text{ when } Sc$
- $Sc, D \not\models \text{necessary alive at } 4 \text{ when } Sc$
- $Sc, D \models \text{possibly alive at } 4 \text{ when } Sc$
- $Sc, D \models \text{possibly } \neg \text{alive at } 4 \text{ when } Sc.$

Thank you for your attention!

Questions are welcome.